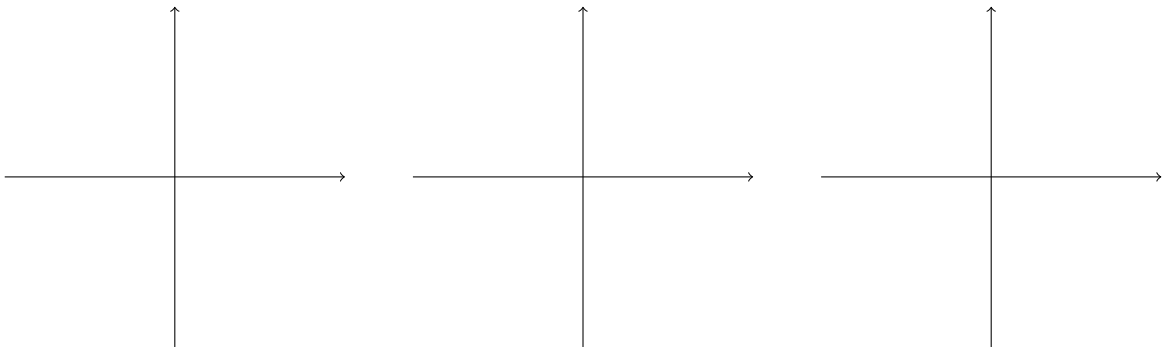
Reference Triangles

For an angle θ in each of the second, third, or fourth quadrants, there is a corresponding angle in the first quadrant, defining a *reference triangle*. You can use the latter to calculate values of each of the trig functions for the angle θ.

On each of the three sets of axes below, draw a unit circle and an angle in quadrant 2, 3, and 4 respectively. Show the corresponding reference triangles in the first quadrant.



Using these, decide whether each of the three trig functions takes positive values and which takes negative value in each of the four quadrants. In the following table, put '+' or '-' in each of the nine spaces.

	$\sin\theta$	$\cos\theta$	$\tan\theta$
2nd quad			
3rd quad			
4th quad			

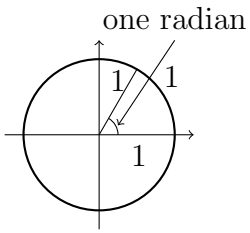
More Special Values of the Trig Functions

Using your known values of sin, cos, and tan for the angle 0°, 30°, 45°, 60°, and 90° and the idea of reference triangles above, find the values of each of the three functions for all of the following angles: 120°, 135°, 150°, 180°, 210°, 225°, 240°, 270°, 300°, 315° and 330°. Show your work. Fill in these angles and the corresponding blank spaces on the unit circle drawn a couple of pages down.

Radians

For reasons that will become apparent next week, degrees are not a sensible measure of angles for any work with trig that involves calculus. Instead, we will measure angles in *radians*.

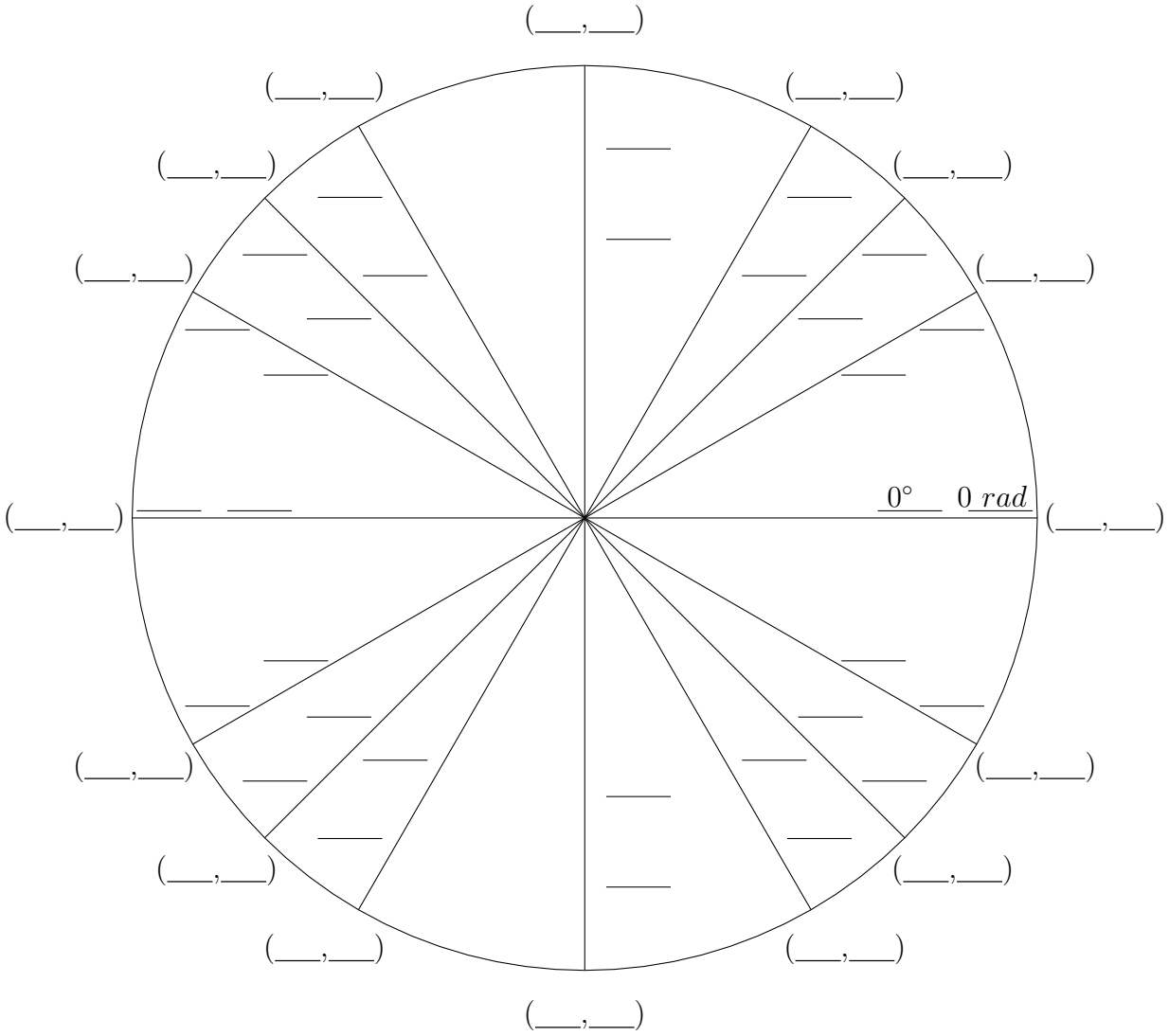
Definition - Radian: One *radian* is the angle in the unit circle at which the corresponding arc-length is exactly 1:



- What is the circumference of the unit circle?
- If the arc-length is 1, what fraction is the arc as part of the entire circle?
- If the angle is θ°, what fraction of the whole circle is that?
- What can you say about your last two answers, and why?
- Use your answer to the last question to find the size of one radian in degrees.
- Lastly, fill on all the radian measures of the angles on the unit circle on the attached page.

- One radian is $\left(_____\right)^\circ \approx$ _____.
- $1^\circ = \left(_____\right)$ radians $\approx$ _____ radians.

The Unit Circle



Homework Problems

Solving Triangles - Trigonometry Angle Problems

Unless otherwise specified, all the below are in radians.

1. For each value of θ, draw the unit circle, the angle θ, and the appropriate "reference triangle". Then use this to determine the *exact* values of $\sin\theta$ and $\cos\theta$: □Regrade

(a) $\theta = \frac{5\pi}{6}$

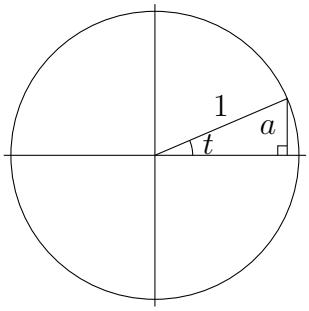
(b) $\theta = -\frac{\pi}{4}$

(c) $\theta = 480^\circ$

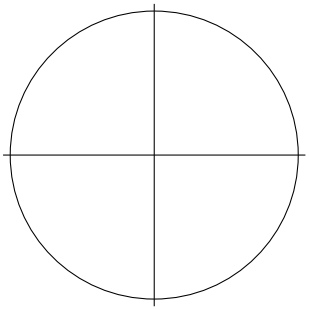
(d) $\theta = \frac{\pi}{3} + 1,000,000\pi$

2. Let t be an angle in the first quadrant, as pictured below. For each of these, draw the given angle and a corresponding triangle congruent to the one given. Then use that to evaluate the given expression in terms of a.

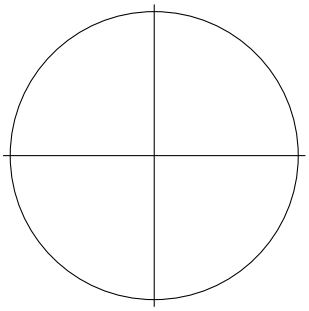
□Regrade



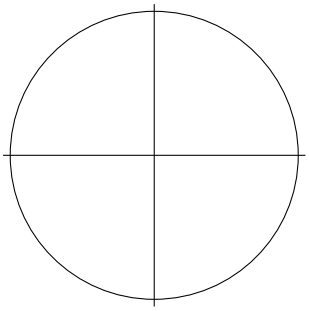
(a) $\sin(t + 2\pi) =$ ____



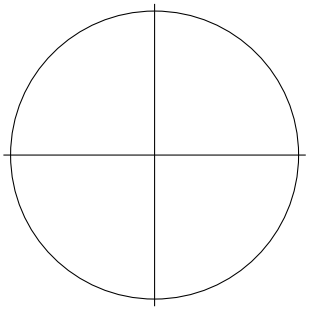
(b) $\sin(t + \pi) =$ _____



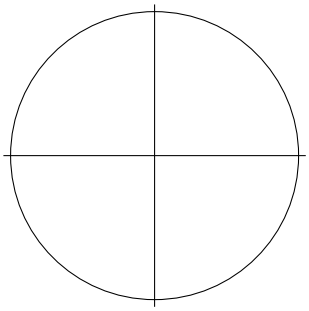
(c) $\cos\left(\frac{\pi}{2} - t\right) =$ ____



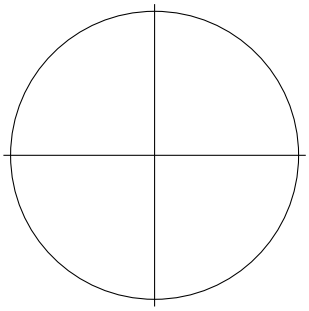
(d) $\sin(\pi - t) =$ ____



(e) $\sin(2\pi - t) =$ _____

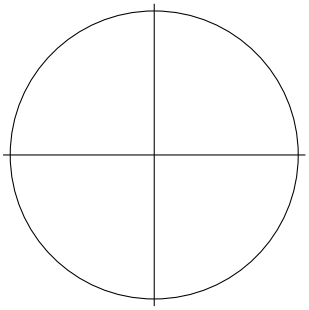


(f) $\sin\left(\frac{3\pi}{2} - t\right) =$ _____

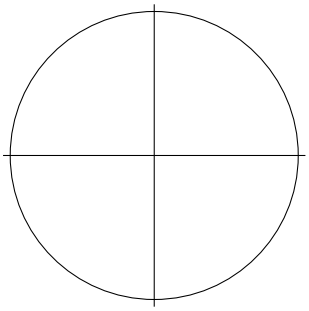


3. For each of these, draw the coordinate on the unit circle corresponding to the given □Regrade angle. Then find its x and y coordinates. If possible, find these coordinates exactly. Otherwise, approximate them.

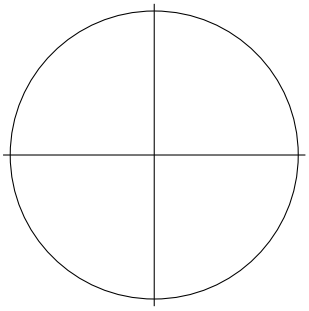
(a) $130°$



(b) $-240°$



(c) $1000°$



4. For what values of θ between 0° and 360° are each of the following true? Explain your □Regrade answers with words and/or drawings.

(a) $\sin(\theta) \geq 0$?

(b) $\cos(\theta) \leq 0$?

(c) $\sin(\theta) \geq \cos(\theta)$?

5. Convert each radian measure to degrees or vice versa. Show work. Exact answers in □Regrade lowest terms only please.

(a) $\frac{\pi}{18}$ radians

(b) -22.5°

(c) $\frac{7\pi}{5}$ radians

(d) 228°.

6. Suppose that the terminal side of an angle θ lies in Quadrant III and lies on the line $\square_{\text{Regrade}}$ $y = 3x$. Draw a unit circle and label this on it. Then use your drawing to find $\sin\theta$ and $\cos\theta$.

(Hint: Draw a right angled triangle!)

7. The Hubble space telescope orbits the earth every 90 minutes in a near circular orbit $\square_{\text{Regrade}}$ 370 miles above the surface of the earth. The radius of the earth is 3960 miles. Write your answers to the nearest mile or hundredth of a radian.

(a) What distance along its orbit will the Hubble telescope travel in 1 hour?

(b) What is the radian measure of the angle through which the Hubble telescope will move after traveling 1000 miles along its orbit? (The angle is measured from the center of the earth.)